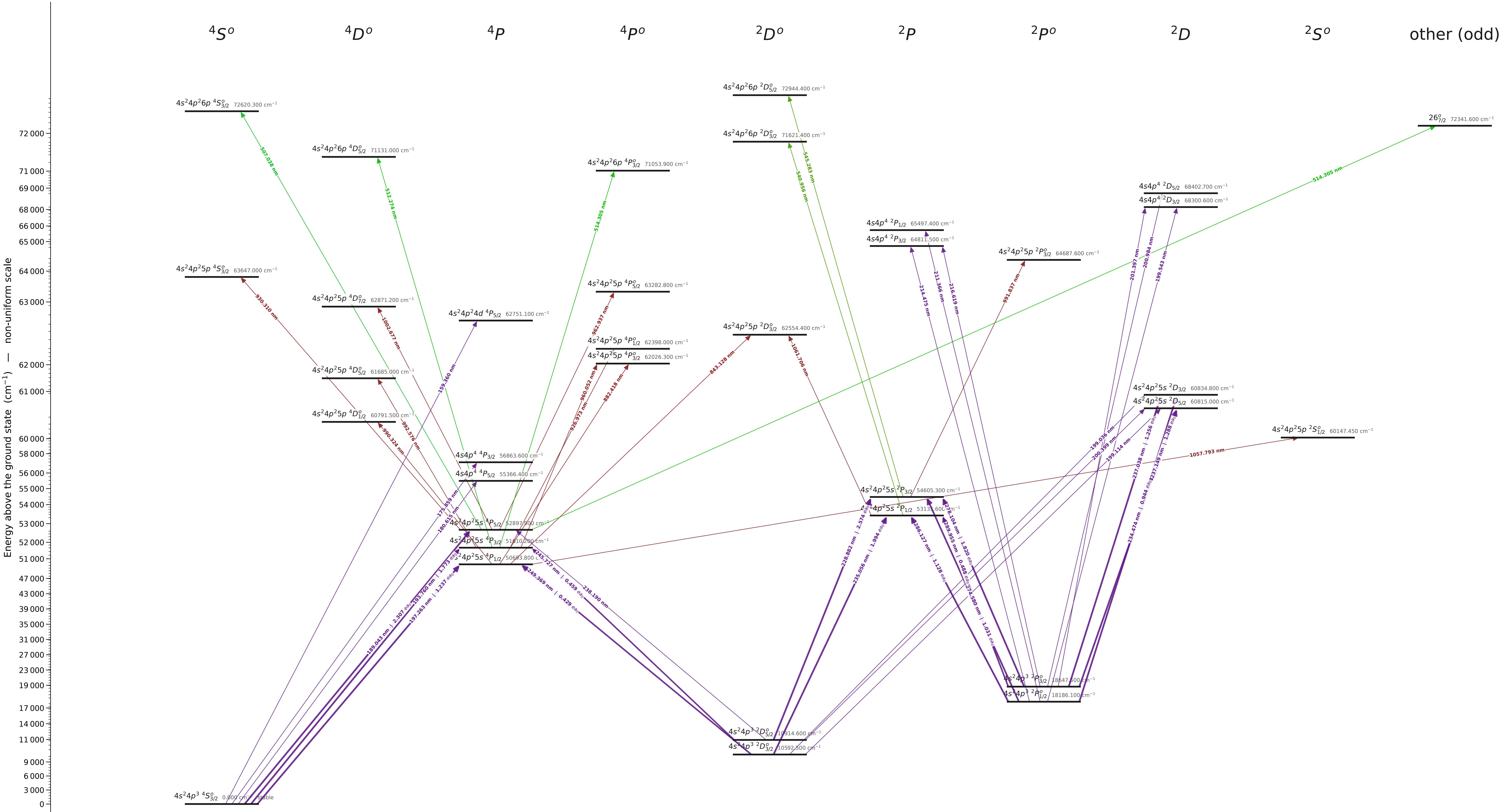
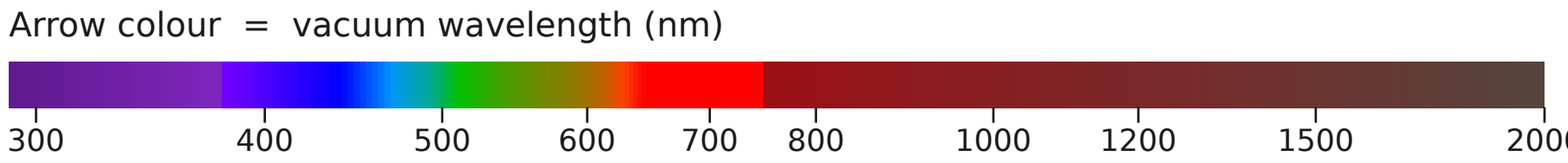


Arsenic

As (As I). Strongest classified transitions below 2 μm (NIST): 35 levels, 45 lines.

As ⁺ ionisation limit 78950.00 cm⁻¹ = 9.78855 eV (126.662 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle U || e r || U \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: NIST energy and measured lifetime (where one exists).



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨U e r U⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
4s ² 4p ³ 4S _{3/2} ^o - 4s ² 4p ² 5s 4P _{5/2}	189.0434	-	1585.8392	2.307	2.660e+08	-	-	
4s ² 4p ³ 4S _{3/2} ^o - 4s ² 4p ² 5s 4P _{3/2}	193.7601	-	1547.2349	1.773	2.190e+08	-	-	
4s ² 4p ³ 4S _{3/2} ^o - 4s ² 4p ² 5s 4P _{1/2}	197.2628	-	1519.7619	1.237	2.020e+08	-	-	
4s ² 4p ³ 2D _{5/2} ^o - 4s ² 4p ² 5s 2P _{3/2}	228.8817	228.8111	1309.8142	2.574	2.800e+08	-	-	
4s ² 4p ³ 2P _{1/2} ^o - 4s ² 4p ² 5s 2D _{3/2}	234.4737	234.4019	1278.5759	0.944	3.500e+07	-	-	
4s ² 4p ³ 2D _{3/2} ^o - 4s ² 4p ² 5s 2P _{1/2}	235.0557	234.9838	1275.4100	1.994	3.100e+08	-	-	
4s ² 4p ³ 2P _{3/2} ^o - 4s ² 4p ² 5s 2D _{3/2}	237.0382	236.9658	1264.7434	1.256	6.000e+07	-	-	
4s ² 4p ³ 2P _{3/2} ^o - 4s ² 4p ² 5s 2D _{5/2}	237.1495	237.0771	1264.1499	1.288	4.200e+07	-	-	
4s ² 4p ³ 2D _{5/2} ^o - 4s ² 4p ² 5s 4P _{3/2}	245.7268	245.6525	1220.0234	0.459	7.200e+06	-	-	
4s ² 4p ³ 2D _{3/2} ^o - 4s ² 4p ² 5s 4P _{1/2}	249.3685	249.2933	1202.2067	0.429	1.200e+07	-	-	
4s ² 4p ³ 2P _{1/2} ^o - 4s ² 4p ² 5s 2P _{3/2}	274.5804	274.4992	1091.8202	1.031	2.600e+07	-	-	
4s ² 4p ³ 2P _{3/2} ^o - 4s ² 4p ² 5s 2P _{3/2}	278.1038	278.0217	1077.9877	1.820	7.800e+07	-	-	
4s ² 4p ³ 2P _{3/2} ^o - 4s ² 4p ² 5s 2P _{1/2}	286.1271	286.0431	1047.7596	1.128	5.500e+07	-	-	
4s ² 4p ³ 2P _{3/2} ^o - 4s ² 4p ² 5s 2P _{1/2}	289.9551	289.8701	1033.9272	0.488	9.900e+06	-	-	

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).